

Project Report: Multivariate Household Electric Power Forecasting

1. Objective

The goal of this project was to perform multivariate time series forecasting on the **UCI Individual Household Electric Power Consumption** dataset. We aimed to predict the `Global_active_power` by comparing classical statistical methods with modern deep learning architectures.

2. Methodology

- **Data Processing:** Aggregated raw minute-by-minute data into hourly averages to reduce noise and capture meaningful patterns.
- **Feature Engineering:** Incorporated multiple variables including reactive power, voltage, intensity, and three sub-metering categories.
- **Stationarity:** Verified stationarity using the Augmented Dickey-Fuller (ADF) test to ensure data suitability for time series modelling.
- **Modelling:** Evaluated four distinct models:
 1. **ARIMA:** A baseline statistical model for non-seasonal time series.
 2. **SARIMA:** An extension of ARIMA that accounts for seasonality (24-hour cycle).
 3. **LSTM:** A Recurrent Neural Network (RNN) designed to capture long-term dependencies.
 4. **TCN (Temporal Convolutional Network):** A convolutional approach utilizing causal and dilated convolutions for sequence modeling.

3. Results Summary

The models were evaluated on a held-out test set (final 15% of the data) using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

- **Deep Learning vs. Classical:** The LSTM and TCN models generally outperformed ARIMA/SARIMA by effectively leveraging the multivariate nature of the dataset and non-linear relationships.
- **Seasonality:** SARIMA showed improvements over ARIMA by accounting for the daily cycles observed during EDA.

4. Key Artefacts

- **Models:** Saved as .pth files in the `models/` directory.

- **Plots:** Visual comparisons of forecasts vs. actual values located in plots/.
- **Data:** Processed metrics and configuration details saved in results/.

5. Conclusion

For this specific power consumption dataset, deep learning models (particularly the TCN) provided the most robust forecasts. Future work could involve hyperparameter optimization or the inclusion of external features like weather data to further improve accuracy.